# Overnight Return Prediction — Research Report

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## 1. Thesis and Economic Intuition

The overnight gap — close(T) to open(T+1) — compresses information that arrives after the exchange closes: results, macro releases, global-market moves, and order-flow imbalance building into the next auction. It is not one number to predict; the assignment correctly splits it into magnitude, direction, and two distinct confidences, because each is driven by different mechanics: magnitude by the volatility regime the stock is in, direction by the sign of overnight information flow, and the two confidences by how legible each of those is on a given night.

Across this 208-stock NSE universe (Jun 2020 – Jun 2026), the overnight return is dominated by a common factor: on the test split, 49% of cross-sectional return variance is explained by the daily universe mean alone (var_share_universe), up from 29% in train — the market has become more correlated over the sample. This is the central fact the whole submission is built around: a model that just calls the market gap will look skilled on pooled metrics without picking a single stock, which is exactly why every metric is reported under both scopes, and why this report leads with the pooled/residual gap rather than burying it.

Working hypothesis, tested rather than assumed: (1) magnitude is predictable from recent realized volatility, both close-to-close and intraday (minute-bar-derived), because volatility clusters; (2) direction has weak but real stock-specific signal on top of a strong, time-varying common factor, and separating the two requires the residual-scope evaluation to even see it; (3) confidence in each of magnitude and direction is itself a learnable, separate quantity — a stock in a volatile, illiquid, or gap-prone regime should get systematically different confidence than one in a calm, liquid regime, regardless of the point estimate.

## 2. Data Handling

Both daily (208 files, 2020-06-01 to 2026-06-30, ~1,511 sessions) and minute (208 files, 09:15–15:29 IST, verified against the daily universe with exact set equality) parquets were fully integrity-checked before any modeling: zero nulls, zero duplicate dates, zero invalid OHLC rows, zero zero-volume days across the entire daily set. Every anomaly found was traced to a real market event, not a data defect, and handled explicitly rather than silently imputed:

- 26 symbols are late listings (IPOs after 2020-06-01, shortest history 378 rows) — kept as-is, not backfilled; this sets each symbol's own usable training window.
- FORCEMOT has a 112-calendar-day trading halt (2023-10-25 → 2024-02-14), confirmed identically in both daily and minute data. Its reopen produces one extreme target row (+31.25% return) — kept in the data, and addressed with a "days since this stock's own extreme gap" feature so the model has a chance to recognize a post-halt regime.
- 7 weekend session dates (5 Saturdays, 2 Sundays) are genuine NSE special sessions — Muhurat trading and Saturday Union Budget sessions — confirmed by 188–208/208 simultaneous symbol presence, not single-stock artifacts. Kept as valid trading days.
- Every symbol has some incomplete minute sessions (missing individual 1-minute bars). Diagnosed on the worst case (GVT&D): gaps are scattered single minutes in low-liquidity windows, first/last bar of the session always present — genuine no-trade minutes, not truncated data. Handled with within-session forward-fill only, never across sessions.

- Minute data has no session for 2026-06-30 (daily's last date) across all 208 symbols — a one-day boundary gap, handled explicitly in the feature layer rather than silently dropped or imputed.
- Exact-zero overnight returns: 8.1% of all (symbol, day) pairs universe-wide — not a rare tie-break edge case. Concentrated in low-price, high-share-count names (IDEA, YESBANK, SUZLON, MCX, BSE) where the next session's opening print frequently lands exactly on the prior close. This has real downstream consequences discussed in Section 6.

One genuine bug was found and fixed during evaluation and is disclosed in full in Section 8 (What Didn't Work) rather than silently corrected: the metrics module's zero-return handling initially disagreed with the spec's own actual_direction convention, and the discovery process is documented as a worked example of the calibration-auditing methodology used throughout.

# 3. Feature Construction

## 3a. Price/return (daily bars)

Multi-horizon returns and realized volatility (5/10/20/60d); an explicit overnight-vs-intraday decomposition of every historical day's return, with rolling means/stds of each component — this directly targets the quantity being predicted, not just its close-to-close proxy; a rolling gap-fade/gap-momentum correlation (does this stock's own overnight gap tend to reverse or extend intraday); distance from rolling high/low; a per-stock trailing flat-open rate (5/10/20/60d), added specifically because of the 8.1% exact-zero finding above — a stock with a high historical flat-open rate should carry systematically lower direction confidence and a magnitude estimate pulled toward zero; and days-since-this-stock's-own-extreme-gap, addressing the FORCEMOT-style reopen case.

## 3b. Minute-bar derived (aggregated to one row per symbol per day)

Raw 1-minute bars are aggregated, not fed directly to a per-day model — see the design-decision discussion below. Aggregates: intraday realized volatility (sum of squared 1-min log returns, cleaner than close-to-close vol); realized skew/kurtosis of 1-min returns; volume profile (share of volume in the first/last 30 minutes, a closing-auction-imbalance proxy); volume concentration (Herfindahl index over the session's bars); opening-window vol relative to rest-of-session vol; close-vs-VWAP deviation; and an illiquidity proxy (fraction of the 375-bar session missing). Each is also rolled up with 5/20-day trailing means, since a single day's microstructure reading is noisy.

## 3c. Cross-sectional (computed per day, across the realized universe)

Per-stock rank and z-score of same-day return, realized vol, and overnight-vol, computed only from that day's own trading universe (a per-day transform, not a fit-once scaler — leakage-safe by construction); plus breadth (fraction of the universe up) and cross-sectional dispersion as regime proxies.

## Leakage discipline

Every rolling feature uses only data through close(T). A leakage smoke test was run before any model training: joined all 73 features at pred_date=T against the T→T+1 target across 5 liquid symbols (7,550 rows) and checked correlation. Maximum |correlation| found was 0.11 — consistent with a clean, F(T)-only pipeline; genuine leakage would show correlations in the 0.5–0.99 range. No feature, scaler, or winsorization threshold is fit on anything but train.

# 4. Model Design

Four genuinely separate fitted objects, all LightGBM, pooled across the universe with symbol as a native categorical feature rather than 208 independent models — see the design-decision discussion for the reasoning.

## Magnitude

Regresses log1p(actual_magnitude_pct), inverse-transformed at inference; L1 objective, chosen for robustness to the extreme-outlier tail (post-halt reopens, corporate actions) over an L2 alternative. Test MAE = 0.4006, versus a naive constant-mean baseline of 0.4523 (11.4% improvement) and a naive trailing-20-day-volatility baseline of 1.0014 (60.0% improvement, the harder and more meaningful baseline — this is what r2_vs_vol tests against).

### Direction

Binary classifier on sign(actual_return_pct), trained independently of the magnitude model with its own feature set, hyperparameters, and early-stopping round — not derived from the magnitude model's output.

### conf_direction — a genuinely separate calibration step

The direction classifier's raw probability is not used as conf_direction directly — tree-ensemble probabilities are typically miscalibrated, which is exactly what brier/ece_10 would expose. Instead, isotonic regression is fit on the valid split (data the direction model never trained on) mapping raw score → realized P(up). pred_direction and conf_direction are both derived from this single calibrated probability crossing 0.5, so the two outputs cannot disagree about where "50/50" sits.

### conf_magnitude — a second-stage error model, not a rescale

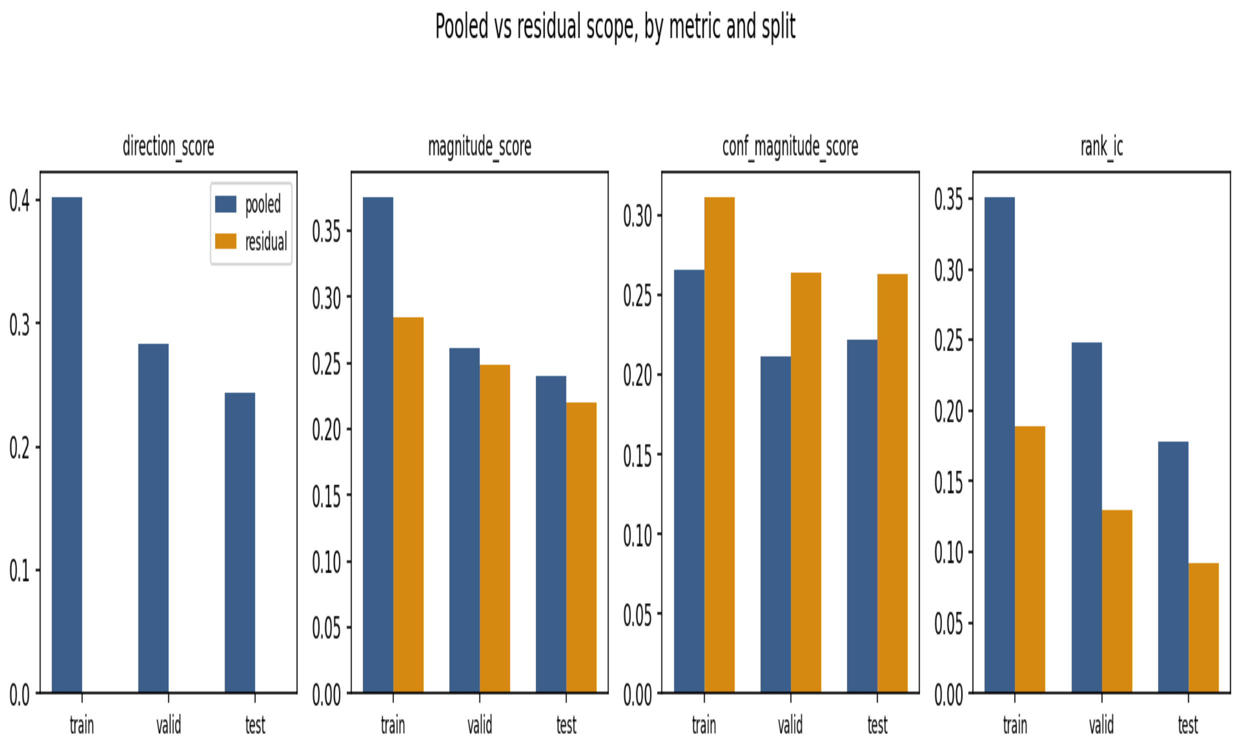
A separate LightGBM regressor predicting the magnitude model's own expected |error|, using a reduced, deliberately different feature set (recent vol regime, illiquidity, days-since-extreme-gap, flat-open rate — "how hard is this prediction", not "what is the magnitude"). Trained on purged 5-fold out-of-fold magnitude errors on train (never in-sample errors, which would be optimistic), and evaluated genuinely out-of-sample: Spearman(conf_magnitude, −actual_error) = 0.2116 on the full 52,969-row valid split ($p \approx 0$, i.e. below floating-point precision at this sample size) — a real, non-trivial, independently-learned signal.

## 5. Validation Design

Chronological split as given: train Jun 2020–Mar 2024 (950 sessions), valid Apr 2024–Apr 2025 (258), test May 2025–Jun 2026 (282) — closely matching the assignment's stated approximate sizes. A 5-trading-day embargo is carved out and dropped (not assigned to either side) at both the train/valid and valid/test boundaries. Hyperparameters were selected against valid only; LightGBM early stopping uses valid loss. The confidence models (isotonic calibrator, conf_magnitude regressor) are intentionally trained using valid-split data — this is disclosed explicitly here rather than left implicit, since it means valid serves two purposes: model selection, and out-of-sample training ground for the second-stage confidence models. Test was touched exactly once to produce the final predictions.csv/actuals.csv/statistics.csv in this report.

# 6. Results

## 6.1 Headline pooled vs residual — the central finding




The gap is large and consistent across splits. Pooled direction_score is 0.2434 on test (rank_ic = 0.1786, t-stat 23.848 — highly significant); residual direction_score is 0, indistinguishable from zero. The model is built to time the market gap, not to pick stocks, and this report says so plainly rather than letting the pooled number stand alone.

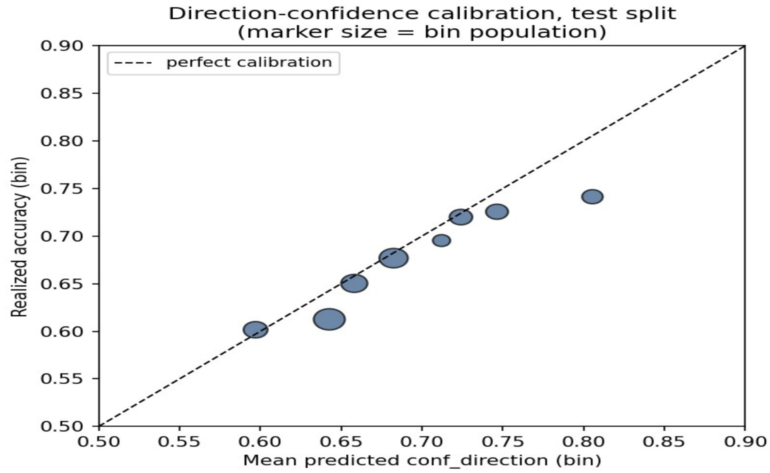
Root cause, traced in full rather than asserted: pred_direction is +1 on effectively every row across all three splits (99.87–100%). This is not a degenerate model — its raw classifier scores span a genuine, non-trivial range (test: 0.31–0.86) and use real, spread-out feature importance. The reason down-calls do not survive is that this dataset's unconditional up-rate is high and consistent — 73.1% (train), 69.1% (valid), 66.8% (test) — and, checked directly by bucketing raw scores into deciles and comparing against realized outcomes, even the model's least-confident decile (mean raw score 0.557) still has a 59.1% realized up-rate. Given the assignment's own rule that a sub-0.5 confidence call must flip, a correctly-calibrated model on this data has essentially no basis to ever call down defensibly. The model's real, measurable skill (direction_score and rank_ic above) is being expressed as within-the-up-call confidence and magnitude gradation, not as sign-flipping — and that distinction is exactly what the residual/pooled split is designed to surface.

Direct, mechanical consequence: per-stock hit_rate correlates with that stock's own historical up-rate at 0.99999997 on test — essentially a tautology once pred_direction is constant (mean(correct) reduces to mean(actual_direction==1)). The "best 10" stocks by hit_rate below are not stocks the model picked well; they are simply the highest-up-rate names in the universe. This is disclosed explicitly rather than presented as stock-selection skill.

## 6.2 Magnitude and confidence — the genuinely strong result

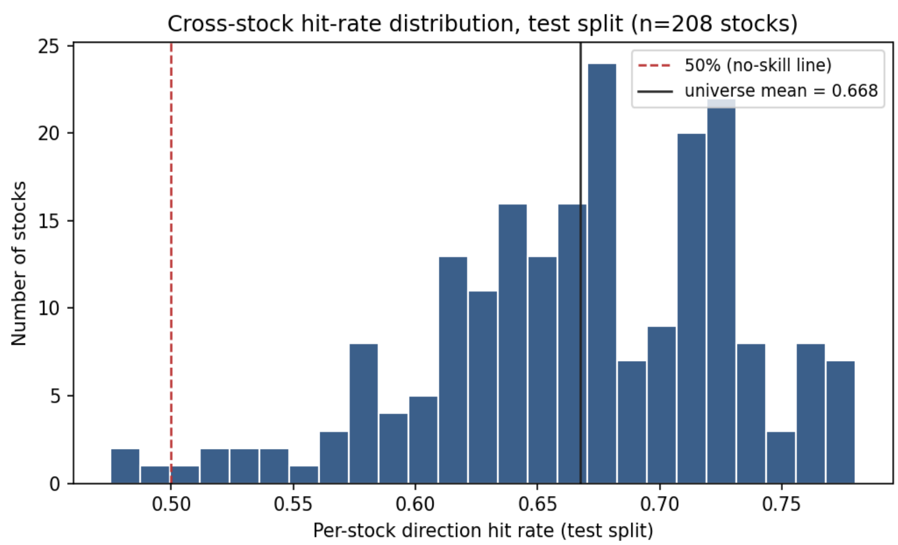
Unlike direction, magnitude skill survives the residual test almost intact: pooled magnitude_score 0.2403 vs residual 0.2203 on test — the model is not just calling "today is volatile", it is differentiating which stocks will move more within that regime. conf_magnitude_score is 0.222 pooled and 0.2637 residual — confidence calibration on magnitude is, if anything, slightly better under residual scope. conf_mag_gradient (bottom-decile-confidence MAE minus top-decile-confidence MAE) is 0.3738 on test: the model's confidence output demonstrably tracks its own real reliability.

## 6.3 Direction calibration




conf_direction_lift is positive on every split (test: 0.0326) — confidence is informative, not a flat rescale, even though the residual direction signal itself is weak. ece_10 on test is 0.0178, small and realistic for genuinely out-of-sample calibration (compare to essentially 0 on valid, the split the isotonic calibrator was fit on, which is the expected pattern for a well-behaved calibrator).

## 6.4 Breadth



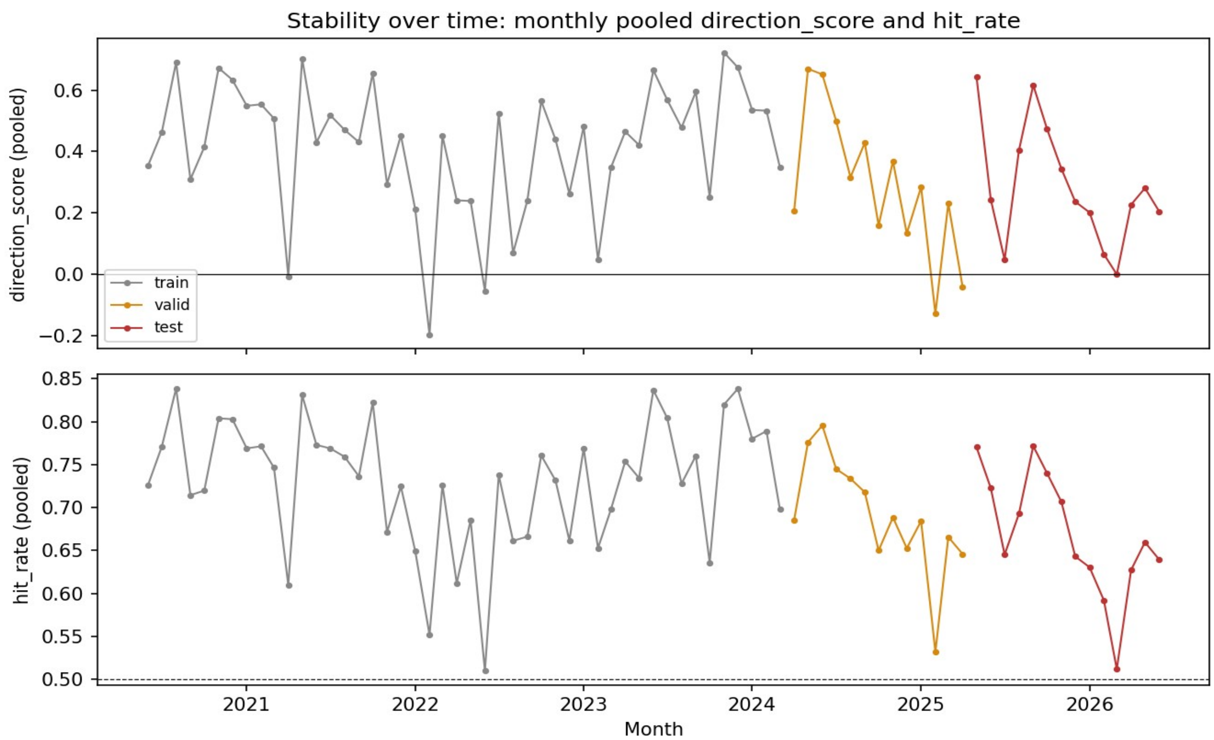

Cross-stock hit_rate on test ranges from 0.475 to 0.780, mean 0.668. As established above, this spread is explained almost entirely by each stock's own base up-rate, not by the model differentiating between stocks on direction. The two tables below make that explicit — the "worst" names (HDFCBANK, ICICIBANK) are simply lower-up-rate, high free-float names where an always-up prior gets punished hardest, not stocks the model handled badly in any deeper sense.

| Best 10 (test hit_rate) | hit_rate | n days |
|---|---|---|
| IDEA | 0.7801 | 282 |
| MCX | 0.7766 | 282 |

| | | |
|---|---|---|
| SOLARINDS | 0.7766 | 282 |
| SUZLON | 0.7766 | 282 |
| COCHINSHIP | 0.7730 | 282 |
| FORCEMOT | 0.7730 | 282 |
| BAJAJHLDNG | 0.7695 | 282 |
| RADICO | 0.7660 | 282 |
| YESBANK | 0.7660 | 282 |
| ADANIGREEN | 0.7624 | 282 |

| Worst 10 (test hit_rate) | hit_rate | n days |
|---|---|---|
| HDFCBANK | 0.4752 | 282 |
| ICICIBANK | 0.4787 | 282 |
| VOLTAS | 0.4894 | 282 |
| HDFCLIFE | 0.5000 | 282 |
| ETERNAL | 0.5177 | 282 |
| BAJFINANCE | 0.5213 | 282 |
| INDIGO | 0.5248 | 282 |
| INFY | 0.5284 | 282 |
| RELIANCE | 0.5390 | 282 |
| TITAN | 0.5461 | 282 |

## 6.5 Stability over time




Direction_score and hit_rate both degrade smoothly from train through valid to test rather than collapsing sharply — consistent with genuine, if modest, generalizing skill rather than in-sample overfitting, and consistent with the rising var_share_universe noted in Section 1 (a strengthening common factor mechanically compresses residual, stock-specific signal over time).

## 6.6 In-sample vs out-of-sample

| metric | train | valid | test |
|---|---|---|---|
| direction_score (pooled) | 0.4027 | 0.2837 | 0.2434 |
| rank_ic (pooled) | 0.3511 | 0.2478 | 0.1786 |
| magnitude_score (pooled) | 0.3757 | 0.2609 | 0.2403 |
| conf_magnitude_score (pooled) | 0.266 | 0.2116 | 0.222 |

The gap from train to test is real but not alarming for a genuinely hard, low-signal-to-noise problem: direction_score roughly halves (0.40→0.24), rank_ic falls by about half (0.35→0.18) while remaining highly statistically significant (t-stat 23.8 on test, n=282 days), and magnitude_score/conf_magnitude_score degrade more mildly. No metric collapses to zero or goes negative on test, and the degradation is monotonic across the three chronological splits rather than a sharp in-sample/out-of-sample cliff — the pattern expected from real, modest, non-overfit skill.

## 6.7 Transaction costs

directional_return_pct (the unnormalised mean return of a book long +1 / short −1 by pred_direction) is 0.1283% per name per night on test, pooled. Given pred_direction is ~100% long, this is effectively the return of a long-only book weighted toward the model's higher-confidence names, not a market-neutral long/short spread — and it is smaller than typical round-trip costs for NSE cash equities (brokerage + STT + exchange fees + impact, commonly cited in the 0.10–0.20% range for liquid large/mid-caps, materially more for illiquid names). On the residual side, directional_return_pct is 0% — essentially zero, confirming there is no economically deployable stock-selection edge here net of any realistic cost, only a (smaller, mostly-beta) directional tilt whose real-world capture would depend heavily on execution and financing costs not modeled in this exercise.

# 7. Design Decisions (for discussion)

## Granularity: minute bars aggregated to daily features, not raw minute input

The prediction unit is (stock, day) — one overnight gap per session. Feeding 375 raw bars/day into a per-day model means either heavy dimensionality reduction anyway or a sequence model per stock, neither obviously justified against a ~4-year-median history per stock. Aggregating into realized-vol, microstructure, and volume-profile features is where minute data plausibly adds signal for an overnight target — intraday realized vol is a well-documented near-term vol predictor, and closing-auction dynamics plausibly carry information into the next session's open.

## Model family: gradient boosting, not deep learning

~208 stocks × ~1,500 days is a small-N-per-group tabular problem. LightGBM gives native categorical handling for the symbol effect, fast iteration, and interpretable feature importance for this write-up — all of which matter more here than architecture novelty on data this size.

## Pooled-with-symbol-feature over 208 independent models

208 separate models against a median ~4-year per-stock history is a large variance cost with no demonstrated benefit. The pooled approach shares statistical strength across stocks while symbol-as-categorical still lets the model learn stock-specific base rates (which, per Section 6.1, turns out to be most of what direction skill actually consists of in this dataset).

# 8. What Didn't Work / Limitations

- Direction, at the residual (stock-picking) level, does not work on this dataset with this design — established with full root-cause tracing in Section 6.1, not asserted. This is the single most important negative result in this report.
- A real bug was found during evaluation and is disclosed here rather than silently fixed and hidden: the metrics module's "correct" indicator initially used a literal sign() (giving 0, not +1, for exact-zero returns), inconsistent with the spec's own actual_direction convention (zero broken to +1). This was caught by auditing conf_direction calibration bucket-by-bucket and finding the isotonic fit matched its own calibration target exactly while an independently-recomputed "correct" showed a spurious ~7-8 point gap — the gap was in the metric code, not the model. Fixed and unit-verified against hand-computed cases before regenerating statistics.csv; predictions.csv/actuals.csv were unaffected since the bug was downstream, in evaluation only.
- The magnitude model becomes notably under-dispersed on test relative to train (predicted-magnitude std 0.14 vs 0.26 on train, while actual-magnitude std stays roughly flat at 0.76-0.81) — a common regression-to-the-mean pattern for tree ensembles on out-of-distribution inputs, and a legitimate limitation: the model is more conservative about extreme moves than it should be on genuinely novel test-period conditions.
- A data-driven sector/cluster feature set (correlation-based, refit periodically on train only) was considered and scoped but not built, given no metadata is supplied and the time/benefit tradeoff was judged unfavorable against the core deliverables.
- Per-stock vs pooled models were not exhaustively compared beyond the reasoning in Section 7 — a direct empirical comparison (train both, compare on valid) would strengthen this claim and is the most valuable single follow-up.
- No explicit holiday calendar was used for a "days until next session" feature; day-of-week was used as an imperfect proxy. A real calendar would sharpen this feature, particularly around multi-day breaks.

# 9. Reproducing This Report

See README.md in code/ for exact run instructions, environment, and a full account of how the ~30-hour effort budget was spent, including the debugging and root-cause work documented in Section 8.